



Research Project

CE-368: Introduction to Reinforcement Learning **Spring 2025**

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1 Introduction

Reinforcement learning (RL) has emerged as a powerful paradigm for training autonomous agents to make sequential decisions by interacting with dynamic environments. Among the most impactful advancements in this field is the development of Deep Q-Networks (DQNs), which combine traditional Q-learning with deep neural networks to enable agents to learn optimal policies directly from high-dimensional sensory inputs, such as images or raw state vectors. Introduced by Mnih et al., [1] the DQN framework successfully demonstrated human-level performance on Atari 2600 games, marking a significant milestone in the application of deep reinforcement learning.

However, the original DQN algorithm exhibited certain limitations, including value overestimation, inefficient exploration, and training instability. In response, several variants such as Double DQN [2], Dueling DQN [3], Prioritized Experience Replay [4], and Rainbow DQN [5] have been proposed to address these issues and enhance learning stability and performance.

This project presents a comparative evaluation of three Deep Q-Network (DQN) variants—Double DQN, Dueling DQN, and Elastic Step DQN—across selected Gymnasium environments with differing dynamics. The study focuses on assessing each method’s sample efficiency, susceptibility to overestimation bias, and convergence behavior using consistent training settings. By leveraging standardized environments such as CartPole-v1 and LunarLander-v2, the project aims to provide practical insights into when and why specific DQN variants may be more effective, with an emphasis on reproducibility and lightweight experimentation.

2 Literature Review

Deep Q-Networks (DQNs) and their variants have significantly shaped the landscape of deep reinforcement learning (DRL), particularly within standardized environments like OpenAI’s Gymnasium. The foundational work by Mnih et al. [1] introduced the original DQN, combining Q-learning with convolutional neural networks, experience replay, and target networks—achieving human-level performance on Atari games. However, it suffered from overestimation bias and high sample complexity. To address this, Van Hasselt et al. [2] proposed Double DQN, which decouples action selection from evaluation, effectively reducing overoptimistic value estimates—demonstrated to improve performance in Gymnasium environments like LunarLander-v2. Complementarily, Wang et al. [3] introduced Dueling DQN, separating value and advantage streams for more efficient learning, particularly in environments with sparse or redundant actions.

In the exploration of Deep Q-Network (DQN) variants applied to Gymnasium environments, Zhong, He, and Kong [6] offer a valuable, though tangential, perspective by examining the integration of Double Q-learning principles within the Asynchronous Advantage Actor-Critic (A3C) [7] framework. The paper proposes two architectures, Double A3C and Less Shared (LS) Double A3C—that aim to reduce overestimation bias by maintaining dual value function estimates (V1 and V2), updating them alternately during training. Although not directly based on DQN, the study aligns with the broader objectives of DQN variants by tackling overestimation and enhancing learning stability. Empirical evaluations across Gymnasium-based Atari games (Breakout, Ice Hockey, and Pong) reveal that while these A3C variants often outperform traditional DQN in terms of average reward and data efficiency, they do not consistently outperform the baseline A3C model, suggesting limited benefit from the added architectural complexity. Particularly in LS Double A3C, reduced parameter sharing leads to potentially noisier updates, and the performance varied significantly by game, highlighting the environment-specific nature of these algorithms. Additionally, the paper emphasizes hardware considerations—such as increased GPU use in asynchronous models versus higher memory requirements in DQN due to experience replay—and calls attention to the trade-offs in resource-constrained scenarios. Despite its limited focus on DQN directly, the paper reflects key trends in deep reinforcement learning: enhancing training stability via architectural innovation,

analyzing algorithm performance across diverse environments, and improving model interpretability through tools like saliency maps. It also underscores the growing challenge of generalization across Gymnasium environments and the importance of tailoring algorithms to game-specific dynamics. These insights highlight gaps in understanding how architectural changes impact stability and overestimation in asynchronous learning settings, which can inform future comparisons between synchronous and asynchronous methods or inspire hybrid approaches that balance sample efficiency with stable value estimation.

Hafiz and Bhat [8] proposes a simple yet effective approach to multi-agent reinforcement learning (MARL), where each agent is a Deep Q-Network (DQN) responsible for a binary action—either to act or not—while sharing a common state and reward signal. This architecture, referred to as Class-Specific DQN (CS-DQN) and its Double DQN variant (CS-DDQN), simplifies the coordination process by removing the need for explicit inter-agent communication and allows the use of shared experience replay buffers. The authors demonstrate the method’s effectiveness on CartPole-v1, LunarLander-v2, and a custom Maze Traversal environment, showing faster convergence and better performance than traditional single-agent DQN/DDQN baselines. A key strength of the approach lies in its simplicity, extensibility to any DQN variant, and resource efficiency. However, its reliance on binary action decomposition and the absence of comparisons with other MARL methods limits its generalizability and broader impact assessment. Moreover, while the shared reward and state structure streamlines learning, it may inhibit the emergence of specialized agent behaviors. The authors identify future work such as introducing inter-agent communication and testing on more complex environments, which could enhance scalability and performance. Overall, this paper contributes a lightweight MARL strategy that performs well in discrete action spaces and aligns with broader trends in reducing complexity in deep reinforcement learning systems.

Exploration remains a central challenge in deep reinforcement learning (DRL), particularly in high-dimensional environments where random exploration strategies are inefficient. Addressing this, Zhang et al. [9] proposed the Noisy Attention Exploration Module (NAEM), a lightweight and generalizable neural architecture designed to enhance exploration through a combination of learnable noise and attention mechanisms. NAEM consists of two parallel sub-modules: a channel noisy attention component, which injects state-dependent Gaussian noise into the attention weights to encourage adaptive exploration, and a spatial attention component that preserves salient spatial features using a CBAM-like mechanism. This design enables effective exploration without disrupting task-relevant feature representations. The authors integrated NAEM into two prominent DRL algorithms, Rainbow (a value-based method) and SAC-Discrete (an actor-critic method) demonstrating broad compatibility and significant benefits. NAEM achieved more than 130% performance improvement over baseline methods (e.g., NoisyNet-DQN, NoisyNet-A3C) on most Atari 2600 games, while also reducing parameters by 44% and speeding up training by approximately 30%. Notably, NAEM offers a novel balance between structured feature extraction and stochastic exploration, outperforming conventional methods such as random action selection or fixed noisy layers. Despite its strengths, limitations include its limited attention to temporal dependencies, which may hinder performance in partially observable or sequential tasks. The authors suggest future work in extending NAEM to recurrent architectures, multi-agent settings, and real-world applications such as robotics, as well as enhancing its interpretability in decision making processes [9].

Ly, Dazeley, Vamplew, *et al.* [10] introduces Elastic Step DQN (ES-DQN), a novel approach that dynamically adjusts the multi-step update horizon in Deep Q-Networks (DQNs) based on state similarity. Instead of using a fixed n -step parameter, ES-DQN leverages unsupervised clustering, specifically HDBSCAN applied to hidden-layer outputs, to group similar states and determine the appropriate update horizon. The method employs a state memory bank to capture diverse representations and refines feature extraction by using hidden node outputs rather than raw states. Empirical results on Cartpole, Mountain Car, and Acrobot demonstrate that ES-DQN achieves competitive or superior performance compared to single-step, fixed n -step, Double DQN, and Average

DQN variants, particularly in reducing the overestimation of Q-values. While the algorithm shows slower convergence during early learning phases and its reliance on clustering may limit scalability to high-dimensional tasks, its dynamic update mechanism effectively alleviates hyper-parameter sensitivity. The study suggests that future work could explore hybrid approaches, alternative similarity measures, and applications to more complex environments, aligning with broader trends in adaptive and resource-efficient reinforcement learning systems.

Skolik, Jerbi, and Dunjko [11] presents a quantum-enhanced deep Q-learning framework that replaces conventional neural networks with parametrized quantum circuits (PQCs) to approximate Q-values as expectation values. The study adapts the DQN architecture by integrating hardware-efficient PQCs and systematically investigates various data encoding and readout strategies—such as data re-uploading and trainable input weights, through detailed ablation studies. The results on benchmarks like Cart Pole show that, while PQC-based agents can achieve competitive performance with fewer parameters, they are notably more sensitive to architectural choices and experience training instabilities as model complexity increases. These findings underline that the success of quantum Q-learning hinges more on careful observable selection and hyperparameter tuning than on mere scaling of parameters. The authors suggest future research directions including improving training stability, developing hybrid quantum–classical architectures, and expanding evaluations to more complex environments, thereby contributing valuable insights toward realizing quantum advantage in reinforcement learning tasks.

3 Research Topic

This project proposes a focused comparative analysis of key Deep Q-Network (DQN) variants, specifically Double DQN, Dueling DQN, and Elastic Step DQN (ES-DQN), across multiple OpenAI Gymnasium environments with varying dynamics and reward structures. While prior research has independently shown the effectiveness of each of these algorithms in certain settings, there is limited empirical work that evaluates them under a common framework to understand their relative strengths, weaknesses, and generalizability across different task types. This gap presents an opportunity to identify patterns in algorithm performance, training stability, and sample efficiency across domains such as dense vs sparse rewards, simple vs complex dynamics, and low vs high variance transitions.

The key research question is:

“How do different DQN variants compare in terms of sample efficiency, overestimation bias, and stability across Gymnasium environments with diverse learning challenges?”

3.1 Justification

DQN-based reinforcement learning algorithms continue to play a foundational role in value-based decision-making systems, particularly for discrete action tasks. The original DQN introduced by Mnih et al. [1] was a landmark contribution but suffered from instability due to overestimation bias and limited generalization. Double DQN [2] addressed this by decoupling action selection and evaluation, while Dueling DQN [3] improved value learning by separating state-value and advantage estimates. More recently, Elastic Step DQN [10] introduced a dynamic update horizon based on state similarity to improve adaptability and reduce hyperparameter sensitivity.

Although these methods address distinct algorithmic limitations, there is little unified analysis across them—particularly in a standardized, low-resource setting such as Gymnasium environments. Much of the existing evaluation focuses on individual algorithm performance on benchmark tasks like Atari games, often with extensive compute budgets. However, practical applications and future

research development benefit from understanding how these methods perform on simpler, faster-to-train environments such as CartPole, LunarLander-v2, and Acrobot.

This project fills that gap by offering a side-by-side comparison under consistent experimental settings. It will evaluate:

- Training sample efficiency (steps to reach a reward threshold)
- Learning stability and variance across seeds
- Q-value trends to infer overestimation or underestimation behavior
- Sensitivity to environment-specific challenges (e.g., sparse rewards or sharp dynamics)

Such a comparison will help clarify when and why to prefer a particular DQN variant, providing useful guidelines for both practitioners and researchers.

3.2 Objectives

The main objectives of this research are:

- To systematically compare Double DQN, Dueling DQN, and ES-DQN across at least three distinct Gymnasium environments
- To evaluate their training efficiency, convergence behavior, and value estimation trends under standardized conditions
- To identify environment-specific strengths and weaknesses of each algorithm

The potential impact of this work lies in its ability to offer actionable insights into DQN algorithm selection for downstream tasks, especially in environments with constrained resources or specific learning challenges. Additionally, the findings may motivate hybrid designs or targeted improvements in reinforcement learning architectures by showcasing complementary behaviors across DQN variants.

By contributing a reproducible and well-controlled comparison of widely used DQN methods, this study aims to enhance practical understanding of deep reinforcement learning algorithms in applied and academic settings.

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